

Calibration Behavior: 2024 vs. 2026

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Overview

This report compares longitudinal calibration behavior between the 2024 and 2026 calibration datasets.

The analysis reproduces the drift analysis originally applied to the 2024 calibration data. For each anchor, calibration is compared with its calibration on the first day of the observation period.

Two measures of drift are examined:

1. **Instantaneous drift:** the absolute proportional difference between an anchor's current calibration and its calibration on Day 0.
2. **Maximum drift:** the largest instantaneous drift observed up to each point in time.

The analysis also examines the number of anchors that exceed 1%, 2%, 5%, 10%, and 20% deviation thresholds.

For the 2026 data, calendar dates are preserved throughout the analysis. Thus, the `days` variable represents actual elapsed calendar days rather than the number of calibration files retained after missing days were removed.

1. Load 2024 Data

The 2024 calibration data are read from individual CSV files. Each filename contains the date of retrieval, which is retained as the observation date, alongside the pre-computed calibration value.

2. Load 2026 Data

The 2026 data requires us to calculate calibration value via given slope and intercept values.

The calibration value is calculated as:

$$y = 100m + b$$

where (m) is the slope and (b) is the intercept.

3. Dataset Summary

- Only anchors with complete observations across all retained calibration dates are used in the final comparison
- The actual calendar dates are retained rather than replacing them with sequential day values

year	anchors	observations	first_date	last_date	calendar_days
2024	659	19111	2024-10-21	2024-11-18	28
2026	444	35520	2026-01-23	2026-04-14	81

4. Drift Analysis

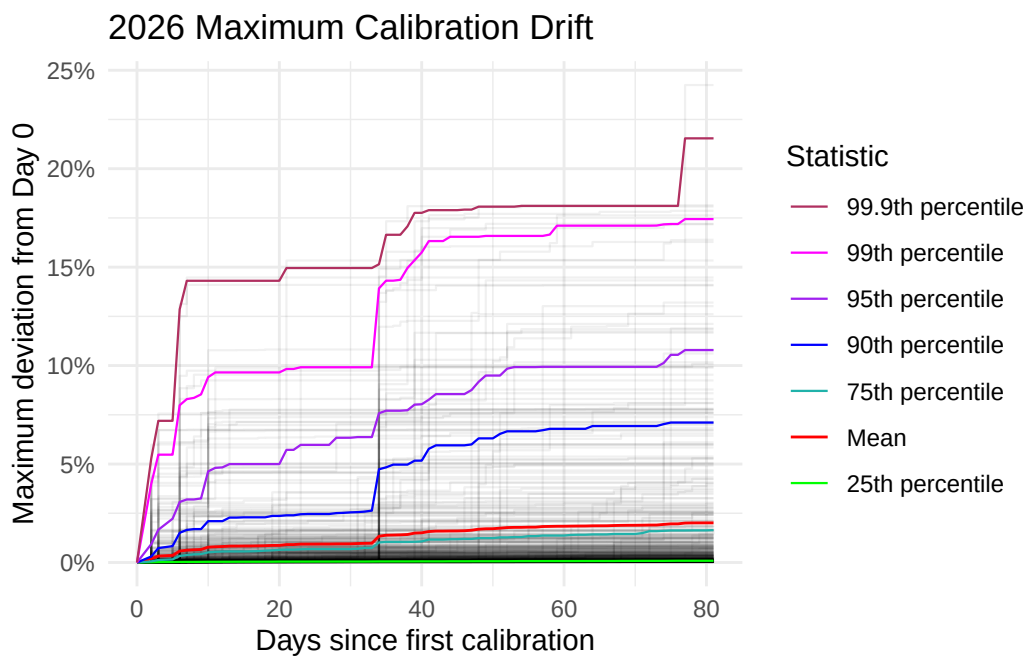
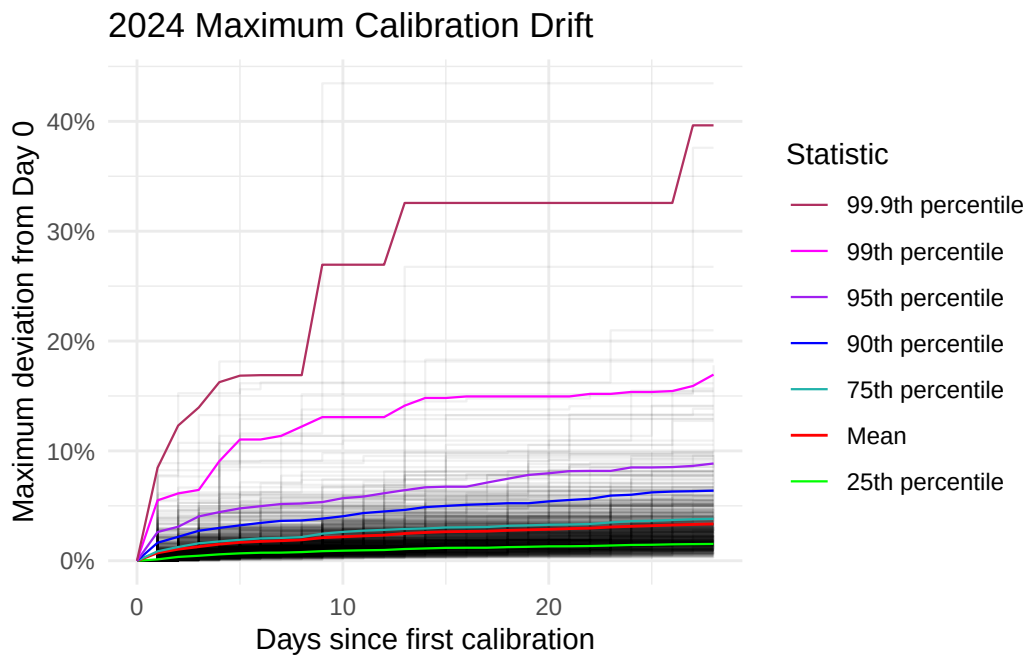
4.1 Calculating Drift

The following calculation is equivalent to Zach's original 2024 analysis.

For each anchor:

$$drift_t = \frac{|calibration_t - calibration_0|}{calibration_0}$$

4.2 Visualizations (1)



4.3 90-99.9th Percentile Anchors

Which anchors are responsible for the region between the 90th and 99.9th percentile at any point during the recorded time period?

- 653 of 659 anchors entered the 90-99.9th percentile range of drift in 2024 data
- 444 of 444 anchors entered the 90-99.9th percentile range of drift in 2026 data

Which anchors were between the 90th and 99.9th percentile on at least once?

(Code for IDs in each category is available in the qmd)

- 653 anchors were in the range at least once in 2024 data
- 444 anchors were in the range at least once in 2026 data
- 276 anchors were in the range at least once in both 2024 and 2026 data
- 185 anchors occupied the range only one time each in both 2024 and 2026 data
- 91 anchors occupied the range more than one time each between 2024 and 2026 data

4.4 Geographic Context of IDs Consistently in the 90-99.9 Range

Below, we take a look at the IDs that appeared in the 90-99.9th percentile range more than one time each between the 2024 and 2026 datasets. These anchors entered the range either once in 2024 and more than once in 2026, or more than once in 2024 and once in 2026, or entered the range more than once in both 2024 and 2026 data. We will discuss the geographic context of these 91 anchors below:

4.4.1 ASN summary

asn	anchor_count	percent_of_anchors
14061	4	4.395604
16276	4	4.395604
3491	3	3.296703
57695	3	3.296703
36236	2	2.197802
197540	2	2.197802

4.4.2 Country Frequency

country	anchor_count	percent_of_anchors
DE	16	17.582418
US	14	15.384615
SG	8	8.791209
AT	7	7.692308
FR	6	6.593407
IT	6	6.593407
CZ	5	5.494505
GB	5	5.494505
NL	3	3.296703
AE	2	2.197802
AU	2	2.197802
HK	2	2.197802
IN	2	2.197802
PL	2	2.197802
SE	2	2.197802

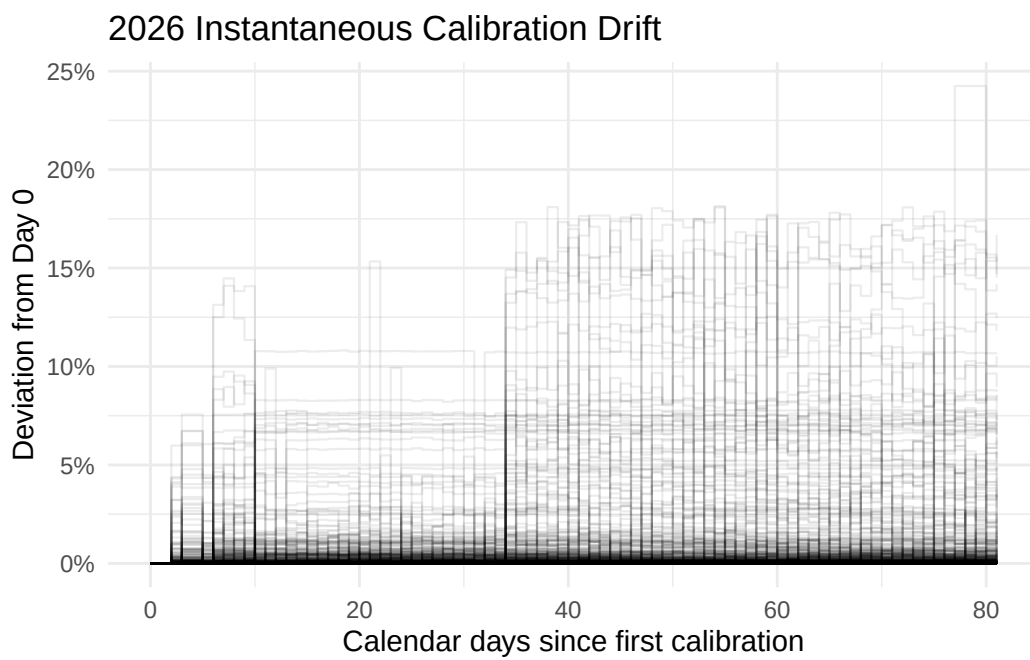
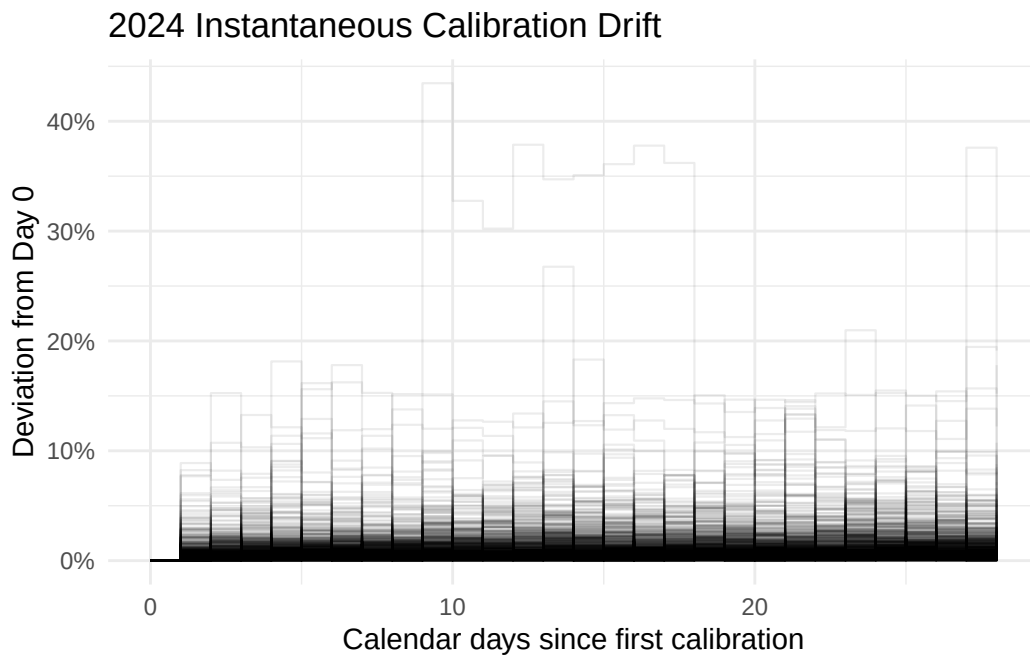
4.4.3 City Frequency

city	anchor_count	percent_of_anchors
Singapore	8	8.791209
Frankfurt	4	4.395604
Prague	4	4.395604
Seattle	4	4.395604
Vienna	4	4.395604
Frankfurt am Main	3	3.296703
Amsterdam	2	2.197802
Berlin	2	2.197802
Dubai	2	2.197802
Hong Kong	2	2.197802
London	2	2.197802
Nuremberg	2	2.197802
Phoenix	2	2.197802
Sydney	2	2.197802

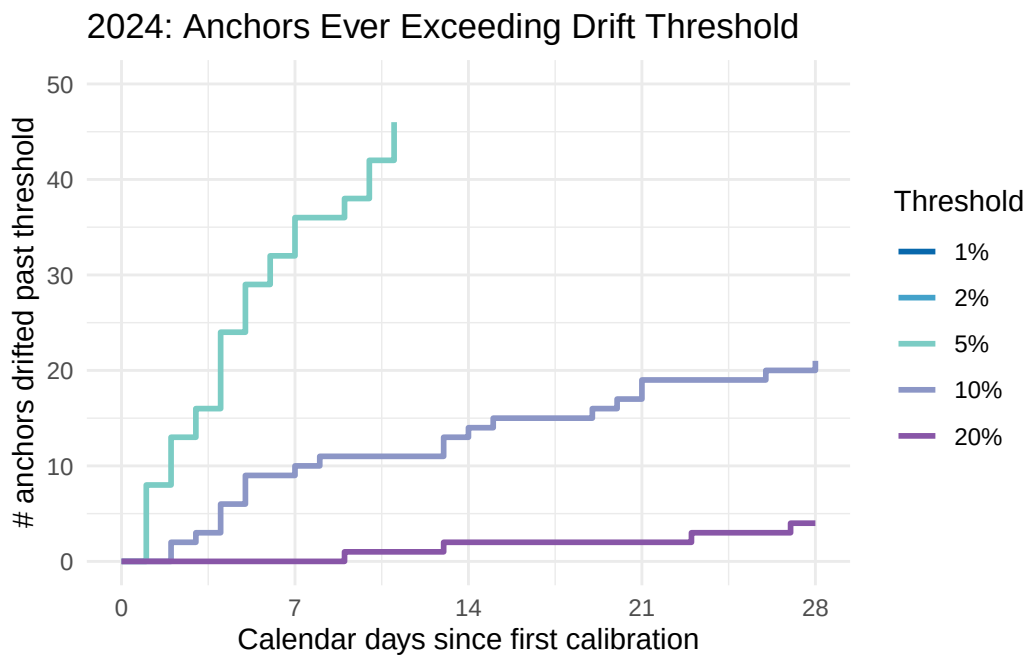
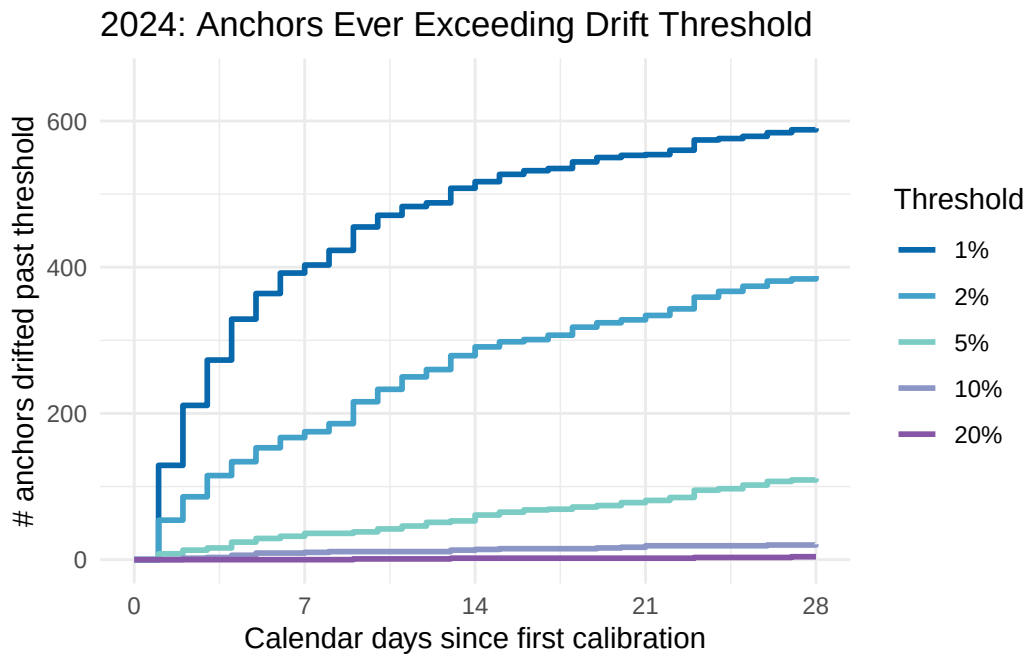
4.4.4 Continent Frequency

continent	anchor_count	percent_of_anchors
Europe	59	64.835165
North America	15	16.483517
Asia	14	15.384615
Oceania	3	3.296703

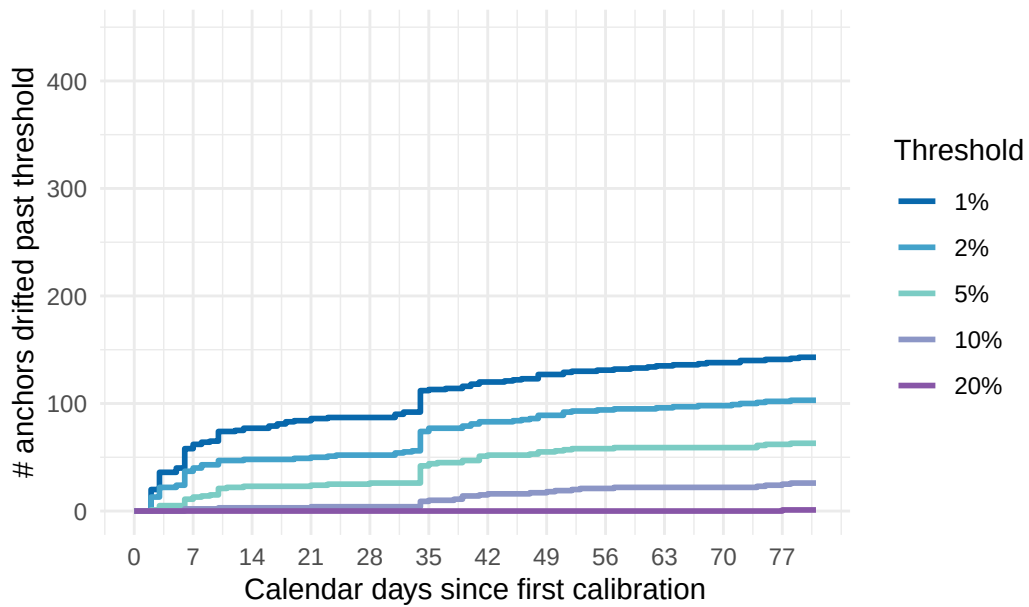
9. Instantaneous Drift



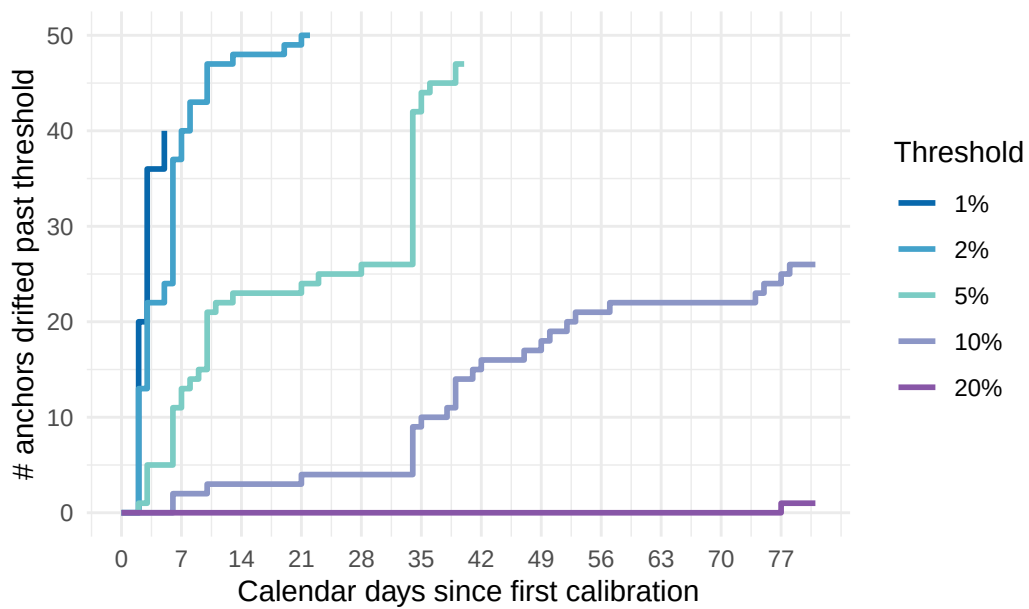
10. Maximum Drift Threshold Analysis



2026: Anchors Ever Exceeding Drift Threshold

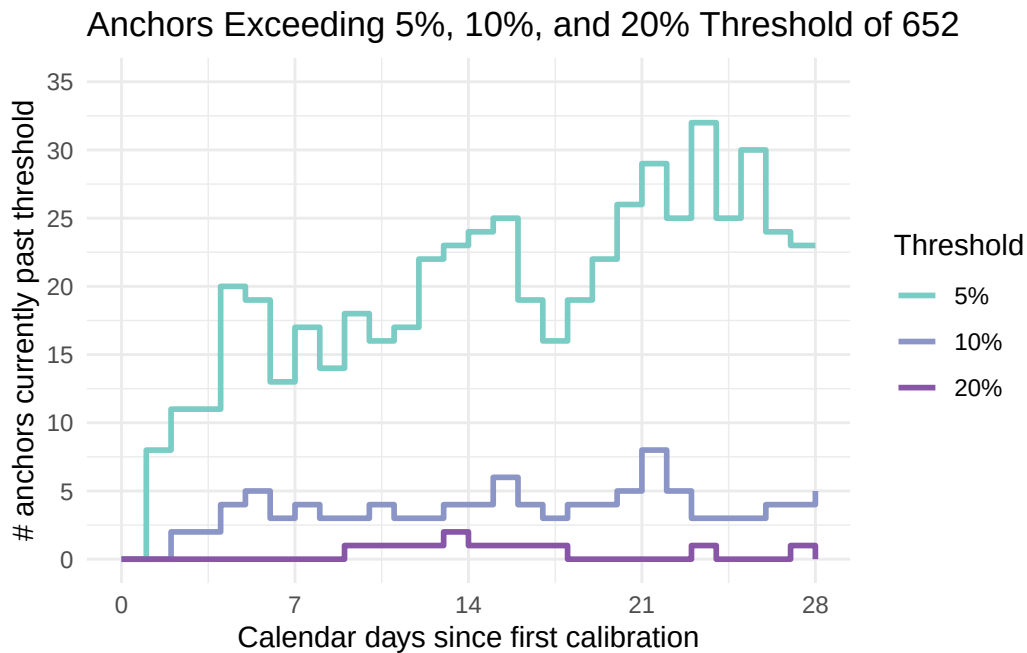
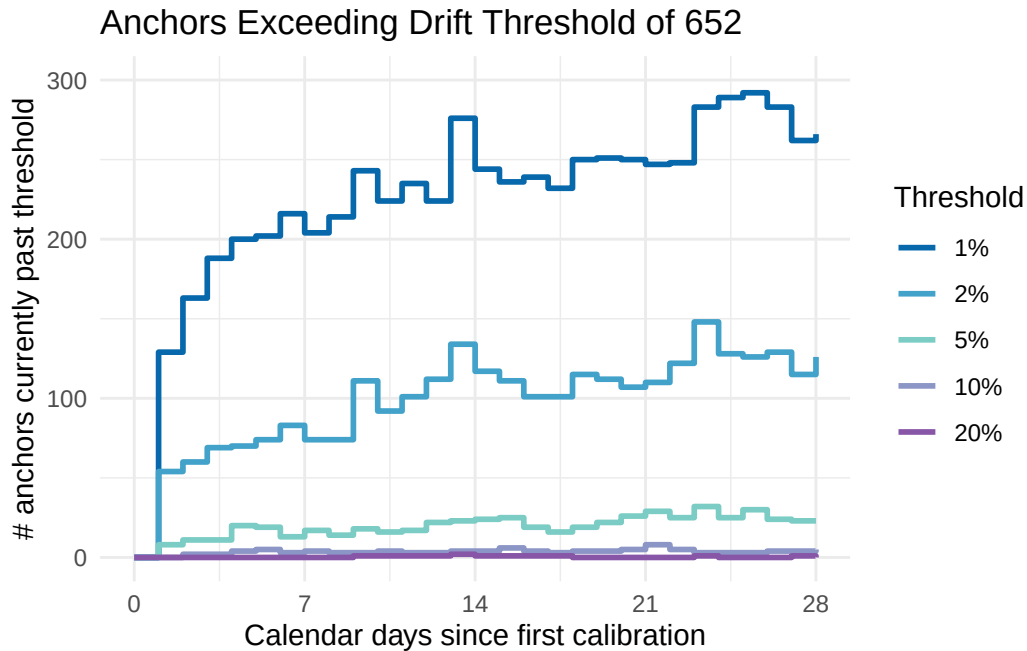


2026: Anchors Ever Exceeding Drift Threshold

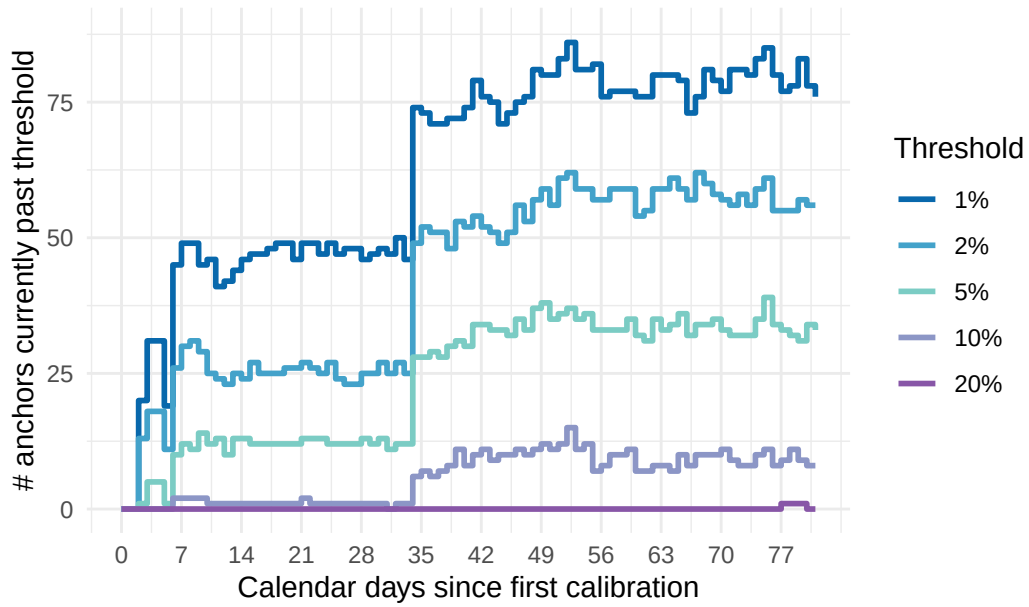


11. Instantaneous Drift Threshold Analysis

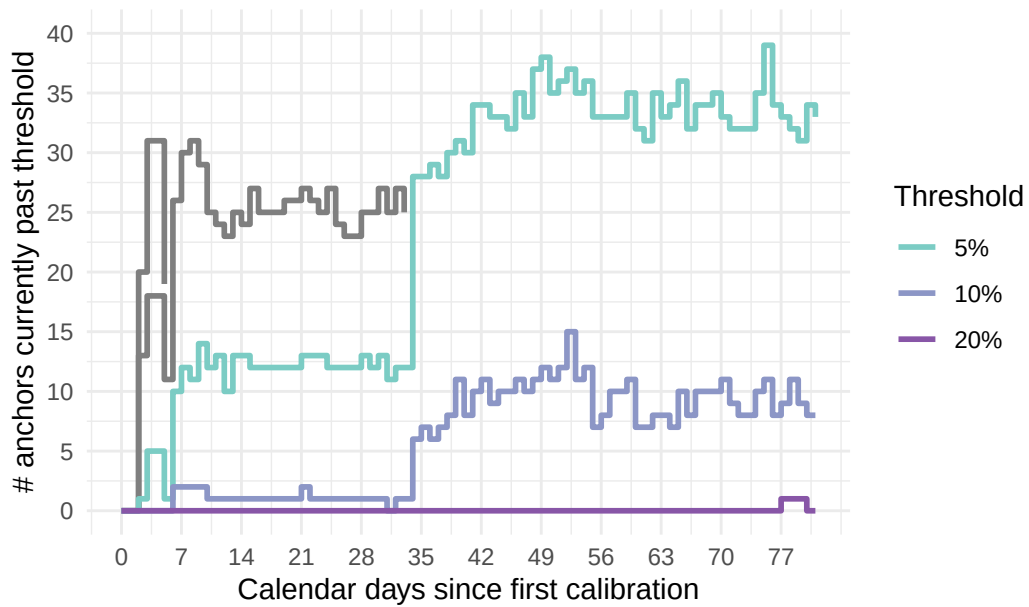
This analysis counts the number of anchors that are currently beyond each threshold, rather than the number that have ever exceeded it.



Anchors Exceeding Drift Threshold of 444

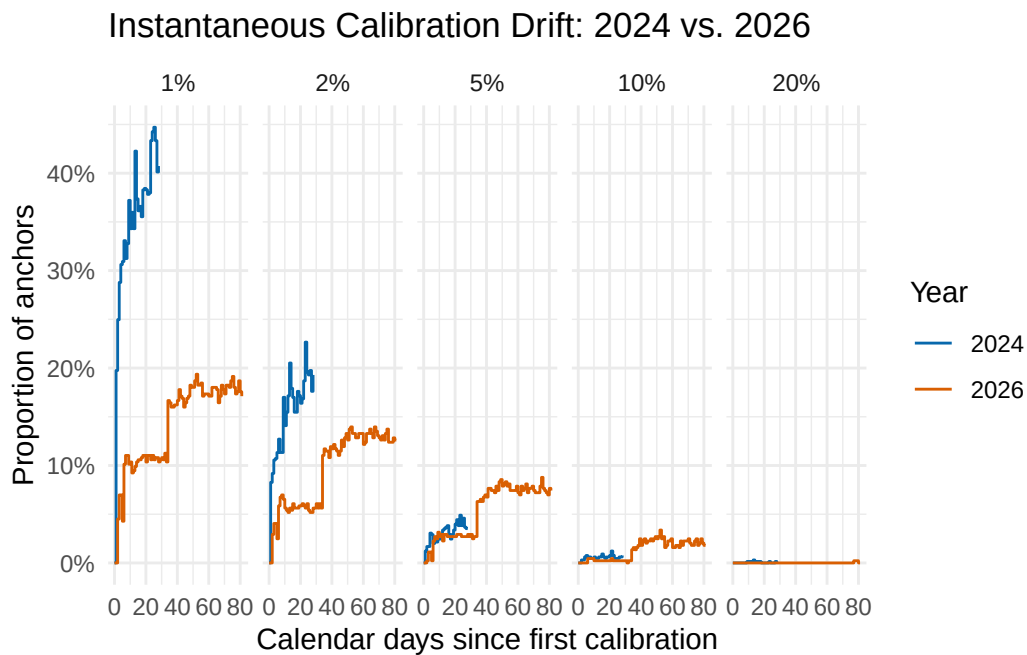
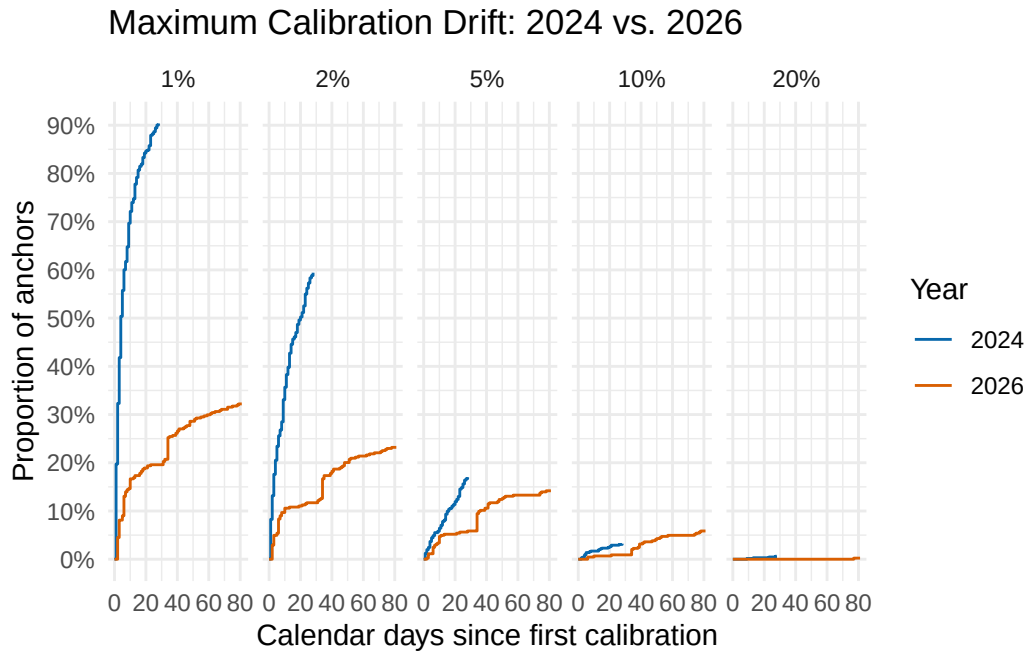


Anchors 5%, 10%, and 20% Threshold of 444



12. Direct 2024 vs. 2026 Comparison

For direct comparison, it is useful to put both years onto the same plot. Because the datasets contain different numbers of anchors, percentages are more informative than raw counts.



14. Notes for Interpretation

The maximum-drift plots measure whether an anchor has **ever** departed from its initial calibration by a given amount. Consequently, these curves can only increase or remain constant.

The instantaneous-drift plots measure how many anchors are **currently** beyond each threshold. These curves can increase or decrease as anchors move back toward their initial calibration.

A pattern in which maximum drift increases over time while instantaneous drift remains approximately stable would suggest that individual anchors experience deviations but that the overall population does not undergo a systematic accumulation of drift.

The 2024 and 2026 comparisons should be interpreted using calendar time. Missing calibration dates in the 2026 dataset remain gaps in the time axis rather than being compressed into consecutive observation numbers.

One thing I'd particularly recommend

The **last two plots** in this QMD are probably the most useful for your actual comparison. They put 2024 and 2026 on the **same percentage scale**, so the fact that you have 444 complete anchors in 2026 versus a different number in 2024 doesn't distort the comparison.

Also, this version fixes an important issue from the earlier approach: **the 2026 x-axis is based on the dates in the filenames all the way through the analysis**. If, for example, 18 calendar days were omitted, those 18 days still occupy space on the x-axis.